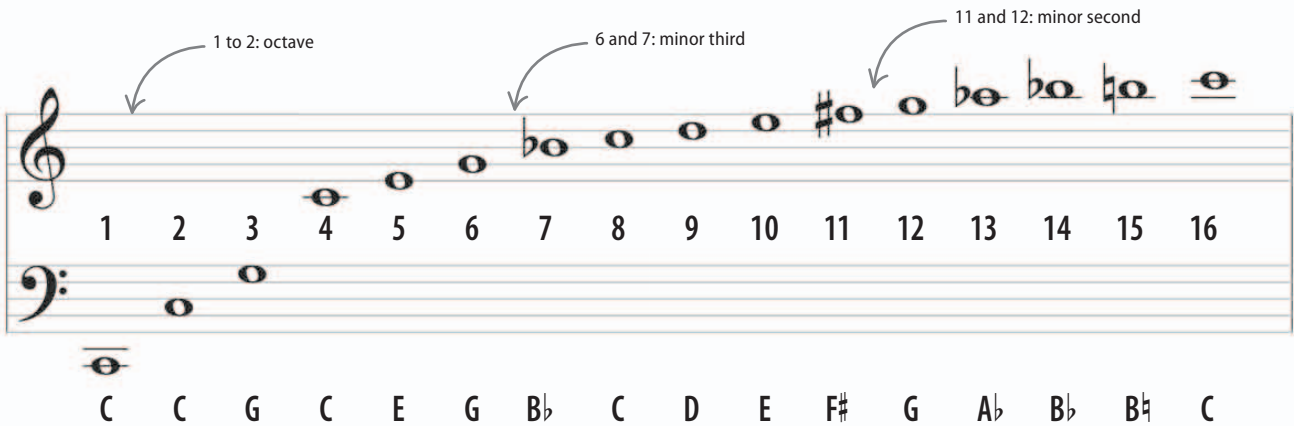


Changing the pitch

A brass player changes notes by increasing the force of the air stream entering the instrument, except with a trombone, or a brass instrument with keys. This is done by loosening or tightening the embouchure. Without keys or valves, the player is limited to the notes of the “harmonic series”—these are the notes played by a brass tube—which get higher in pitch as the air flow in the tube intensifies.

▽ Harmonic series

A harmonic series can be defined as a series of notes that are produced while playing any instrument. The lowest note of this series is called a “fundamental.” The other/higher notes are known as “overtones,” “harmonics,” or “partials.”

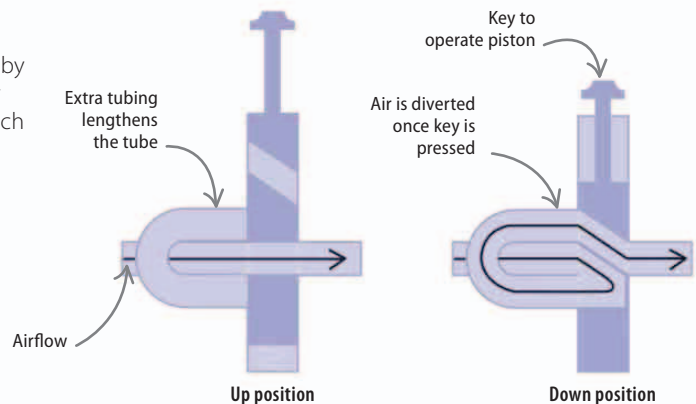


Lengthening the tube

The fundamental note of a brass instrument can be altered by lengthening the tube. This makes the “harmonics,” or higher notes, available. Trombones do this by means of a slide, which can be extended. Trumpets, horns, and tubas use special mechanisms to divert air into extra tubing, temporarily lowering the fundamental note.

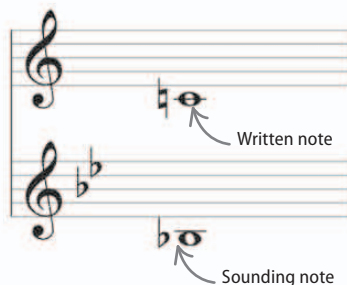
▷ Piston mechanism

Systems of pistons or valves are used to divert the air into the extra tubing. The player presses down a key to operate a piston, and the air is diverted into a special loop, elongating the vibrating length.



Transposing instruments

Instruments that use music written in a different key (called “concert” pitch) from that of the actual sound are called transposing instruments. This distinction between written and sounding notes makes it easier for the player to read the music, and is most often used for brass instruments, clarinets, and saxophones.



△ B $\flat$ trumpet

A written C, when played on a B $\flat$ trumpet, sounds as B $\flat$.

LOOKING CLOSER

Special effects

Brass instruments produce special effects by inserting a metal or wood mute into the bell of the instrument. Straight and cup-shaped mutes are used to produce a quiet, nasal sound.

▷ Mutes

A conical mute deadens the sound of a trumpet or trombone, making its sound less prominent in the ensemble.

